AN INTRODUTION TO REACTOR ANTINEUTRINO FLUXES CHALLENGES

Mehran Dehpour and Sahar Safari

January 2021, Shahid Behshti University

REACTOR NEUTRINO EXPERIMENTS

Production

Prompt neutrons

- Consider a nuclear fission for e.g.

$$n + {}^{235}U \longrightarrow X + Y + \boxed{kn}; where\ k = 2.4$$

- Recall the products of fission may β-decay or release neutrons, these are delayed neutrons

- Thus β-decay of prompt and delayed neutrons can produce $\bar{\nu}_e$ following

$$n \longrightarrow p + e^- + \bar{\nu}_e$$

- Note that flux of antineutrino from reactor is isotropic but we have low energy $\bar{\nu}_e$ and it's hard to detect BG effects and these are SBL almost

Detection

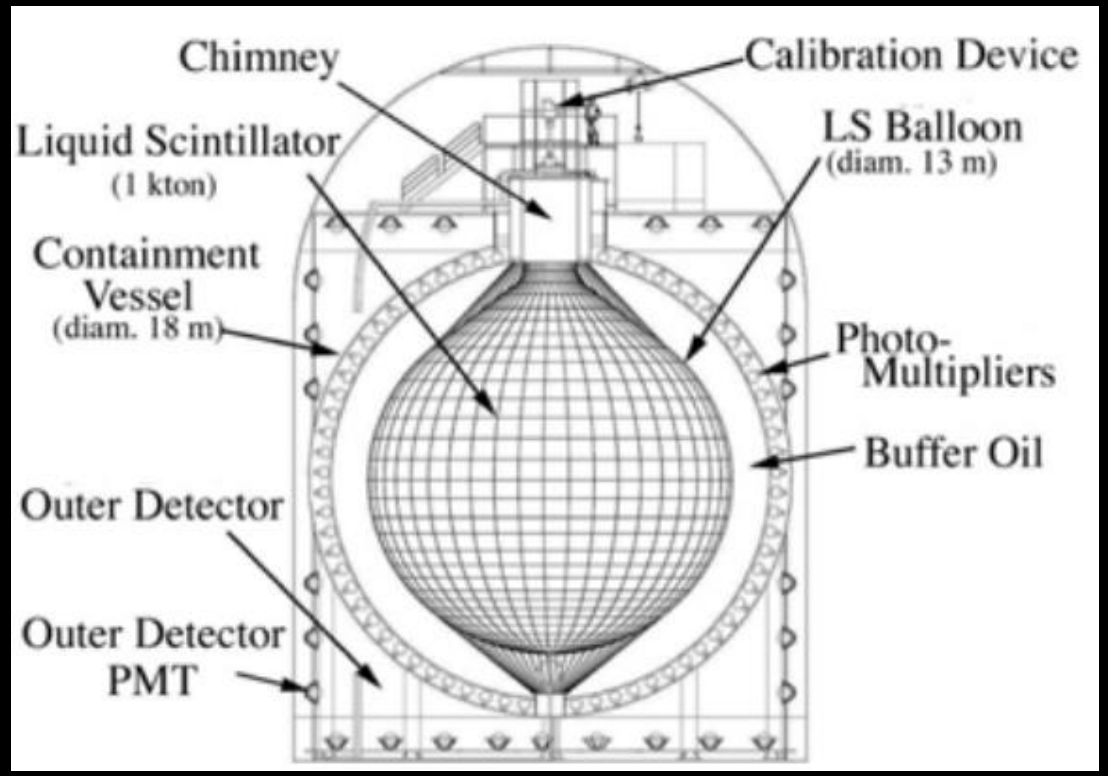
https://doi.org/10.1142/9789813226098_0007

Positron interact with liquid scintillation and release photon — Photons detected by PMTs

- The main parts of a detector are
  - Photon Detector (PMTs)
  - Liquid Scintillation
  - Neutron Poison (Absorber for e.g. Gd)
- So we can detect $\bar{\nu}_e$ via IBD

$$\bar{\nu}_e + p \longrightarrow n + e^+$$

- The reactor antineutrinos are produced at a rate of $\sim 10^{21}$ per s per GW_{th}
- Note that we are must absorb neutrons inside detector
- At last we are corrections by calc Gd β-decay and Cosmic Ray isolate and calibrate detector when reactor off for delete background effects



Schematic diagram of the KamLAND detector

Detection

http://dx.doi.org/10.1093/acprof:oso/9780198508717.001.0001

- According to energy-momentum cons $\bar{\nu}_e$ needed energy to dectection

$$E_{\bar{\nu}_e}^{th} = \frac{(m_n + m_e)^2 - m_p^2}{2m_p}$$

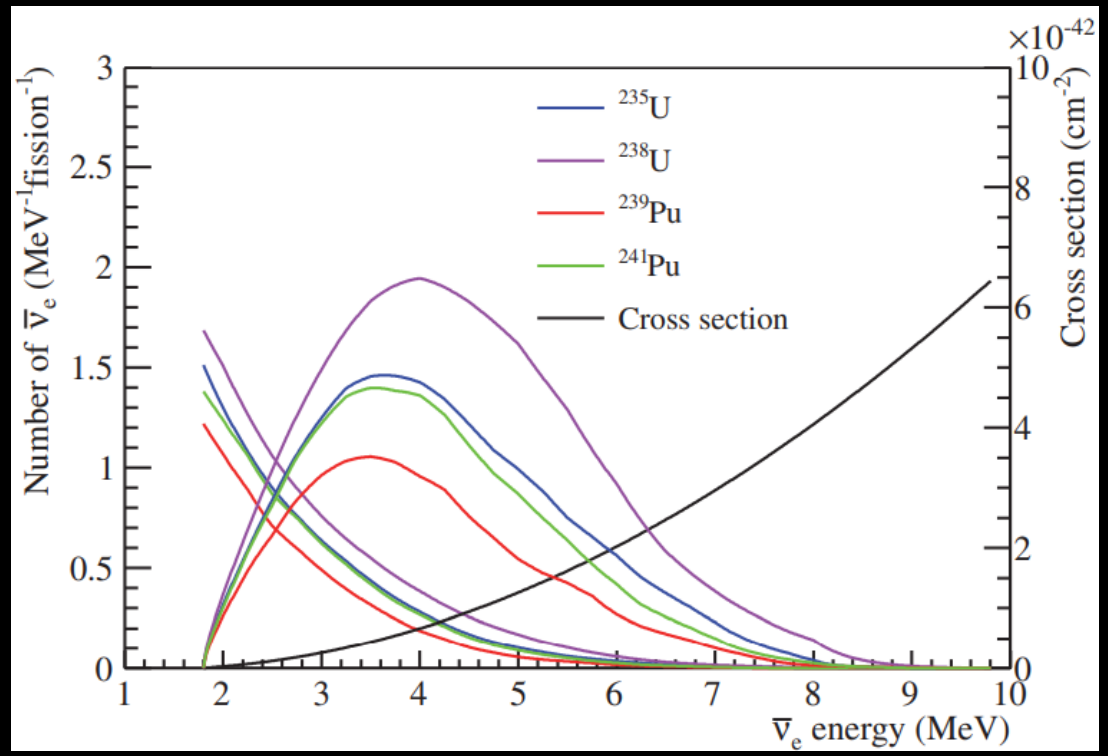
$$\approx 1.806\ MeV$$

that is limit our experiment and lost 75% of data

- Cross-section is geiven by

$$\sigma_{CC}^{\bar{\nu}_e p} = \frac{G_F^2 |V_{ud}|^2}{\pi}(g_V^2 + 3g_A^2)E_p p_e$$

$$where\ g_V \approx 1\ and\ g_A = 1.27$$



Sensitivity

http://dx.doi.org/10.1093/acprof:oso/9780198508717.001.0001

- Sensitivity of this experiments dependent to
  - Base line
  - Energy of $\bar{\nu}_e$
  - Power of source
  - Mass of detector
  - Detector cross-section
  - Background effects
  - Neutron Poison
  - …
- Experiment Conditions
  - SBL: $E \sim 1\ MeV,\ L \sim 10\ m,\quad \Delta m^2\ sensivity \sim 0.1\quad eV^2$
  - LBL: $E \sim 1\ MeV,\ L \sim 1\ km,\quad \Delta m^2\ sensivity \sim 10^{-3} eV^2$
  - VBL: $E \sim 1\ MeV,\ L \sim 10^2\ km, \Delta m^2\ sensivity \sim 10^{-5} eV^2$

- Some of Detectors
  - ILL, Gosgen, Savannah River, Rovno, Krasnoyarsk and Bugey
  - CHOOZ, Palo Verde and KamLAND
  - Double CHOOZ, Daya Bay and RENO

Aims

Determined by atm and solar nu exp

Determined by reactor LBL exp at RENO on 2012; $\theta_{13} \approx 8.32°$

- Determine oscillation parameters (PMNS matrix elements)
  - Two parameters for two nu osc (a mixing angle and squared mass difference)
  - Four parameters for three nu osc (θ_{12}, θ_{32}, θ_{13} and a phase angle)

  In three-flavor oscillation model, the survival probability

  $$P(\bar{\nu}_e \longrightarrow \bar{\nu}_e) \approx 1 - cos^4\theta_{13} sin^2 2\theta_{12} sin^2\Delta_{21} - sin^2 2\theta_{13} sin^2\Delta_{ee}$$

  Note that we can't measured phase angle with this experiments

  This oscillation probability does not depend on the CP phase

- Monitoring activity inside the reactor for reducing threats and preventing the proliferation of nuclear weapons

So we need have ability to accurately predict neutrino flux for this aims

NEUTRINO FLUX

Determine Neutrino Flux, N_υ

- Determine N_υ directly by near detector site
  - At osc exp use this way today but most of the time this method is not possible
- Determine N_υ by ‘ab initio’ summation
- Determine N_υ by phenomenological methods (electron spectrum conversion methods)

Fermi Model For β-decay

- Consider
  - Electron and neutrino does not exist from beginning and made in nuclear
  - Electron and neutrino are relativistic
  - There must be a continuous distribution of electrons
- Use Fermi Golden rule

$$N_\beta(\mathrm{W}) = \mathrm{K}\, p^2 (\mathrm{W} - W_0)^2$$

Fermi Model Corrections

arXiv:1605.02047
arXiv:1106.0687

- Effect of nuclear Coulombic force on electrons (Fermi function)
- Finite size corrections
  - EM
  - Weak Interaction
- Screening correction
- Radiative corrections
- Recoil effect
- Weak magnetism

We can neglect this effects

$$N_\beta(\mathrm{W},\mathrm{Z}) = \mathrm{K}\, p^2(\mathrm{W} - W_0)^2\, F(Z,W)\, L_0(\mathrm{Z},\mathrm{W})\, \mathrm{C}(\mathrm{Z},\mathrm{W})\, \mathrm{S}(\mathrm{Z},\mathrm{W})\, G_\beta(\mathrm{Z},\mathrm{W})\, (1 + \delta_{WM} W)$$

‘ab initio’ summation

arXiv:1902.03281
arXiv:1605.02047

- Related on evolution of nucleus in the reactor
- This method need more data and info about nucleus and decay branches and we may lose some data

$$\frac{dN_i}{dE_\nu} = \sum_n Y_n(Z, A, t) \sum_{n,i} b_{n.i}(E_0^i)\, N_\beta\left(E_\nu, E_0^i, \mathrm{Z}\right)$$

‘ab initio’ summation e.g.

[hep-ph] 6 May 2016

Reactor Neutrino Spectra

Anna C. Hayes,[1] and Petr Vogel[2]

[1]T-2 Theoretical Division MS283, Los Alamos National Laboratory, Los Alamos, NM 8545, USA ; email: anna_hayes@lanl.gov
[2]Kellogg Radiation Laboratory 106-38, California Institute of Technology, Pasadena, CA 91125, USA ; email: pvogel@caltech.edu

Phenomenological method

arXiv:1106.0687
arXiv:1101.2663

arXiv:1312.5601
ILL papers (for e.g. Schreckenbach1985)

- Using β-decay spectra

$$N_\upsilon(E) = N_\beta(E - 511keV - 50keV)k(E)$$

- Another hand we know

Vogel1989

$$\frac{dN_\upsilon}{dE_\nu} = \exp(a_0 + a_1 E_\nu + a_2 E_\nu^2)$$

arXiv:0407026
arXiv:1101.2755

Phenomenological method e.g.

Volume 160B, number 4,5 PHYSICS LETTERS 10 October 1985

DETERMINATION OF THE ANTINEUTRINO SPECTRUM FROM ^{235}U THERMAL NEUTRON FISSION PRODUCTS UP TO 9.5 MeV

K. SCHRECKENBACH [a], G. COLVIN [a], W. GELLETLY [b] and F. VON FEILITZSCH [c]

[a] *Institut Laue Langevin, 38042 Grenoble, France*
[b] *University of Manchester, Manchester M13 9PL, England*
[c] *Physik Department, Technische Universität München, 8046 Garching, West Germany*

Received 17 July 1985

ANOMALIES

Reactor Anomaly

arXiv:1602.01499
https://doi.org/10.1142/9789813226098_0007

- ILL predict with 2% err N_υ
- Double CHOOZ find new err, so we have 3% err in N_υ
- Correction in neutron lifetime, improve err to 6% —— This is seen in Daya Bay
- By multiplying a fixed number by the value of the theory can make it consistent with observation

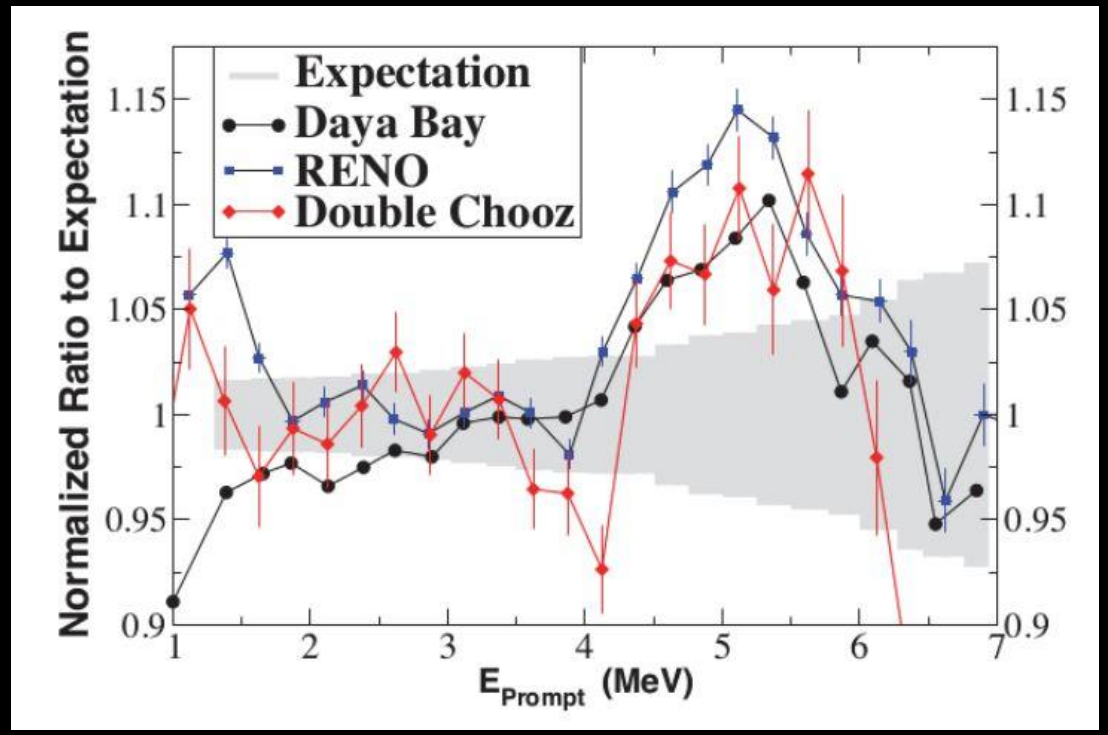
Origins of the Reactor Anomaly

- For basic neutrino research, the question is whether the 6% deficit is
  1. Due to Nuclear Physics
  2. Or due Neutrino Oscillation Involving One Or Several eV-scale Sterile Neutrinos

5 MeV Anomaly

arXiv:1602.01499
https://doi.org/10.1142/9789813226098_0007

- RENO, Daya Bay and Double CHOOZ see a bump between 4.8-7.3 MeV!



The expectation region in the Fig comes from Huber and Mueller`s Phenomenological methods

Origins of the Bump (5 MeV Anomaly) arXiv:1506.00583

- Hayes and collaborators investigate the possible origins of the observed shoulder and propose the following list:
  1. Beta decay of non-fissionable material in the reactor
  2. Shape of the beta and neutrino spectrum for $\Delta J^{\Delta\pi} = 0^-$ first forbidden decays
  3. Beta decay of the daughters of the fast fission of ^{238}U
  4. Beta decay of daughters of the epithermal fission of ^{235}U, ^{239}Pu and ^{241}Pu
  5. Errors in Schreckenbach's ILL beta spectra

THANKS FOR YOUR ATTENTION